MONISHA JEGADEESAN

Senior Software Engineer, Google, New York

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With over six years of experience building scalable client infrastructure across Google Sheets, Android Keep, and Google Docs; a Master's degree in Computer Science; and a research background in NLP and deep transformer models, I specialize in infrastructure, data transfer, performance, and memory optimization techniques, including data compression, network syncing, and on-device machine learning models - for crash-resistant systems used by millions.

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Google Sheets Web Core Infrastructure | Google LLC, New York ***Jan 2025 - Present***

- Drove the multi-stage launch of a data compression infrastructure, enabling support for **large sheets with up to 20M cells**, improving serialized document size, as well as latency of **writing/reading from storage by up to 8%**
- Owned the rigorous data representation accuracy effort, enabling rollout of the **proto data compression** infrastructure**, with up to 10% improvement in initial load and up to 40% improvement in sheet copy.**
- Engineered layout and data capture improvements targeting user journeys such as filtering and duplicating data, leading to improvements of **~1.5% in heap memory,** ~**4% in crash rate,** and **10% in the edit latency.**

Software Engineer | Google Keep Android Notetaking App | Google LLC, New York ***Sep 2022 - Dec 2024***

- Led the infrastructure effort to sync image media data between the Android client and the Java server for a new type of note in Google Keep, with a focus on **image insertion with a close to 0% upload error rate.**
- Designed techniques for optimal data storage and transfer: dynamic batching for network syncing, chunked local storage, optimal serialization format, etc - improving **transfer time by 33%** and cutting down **size by 80%.**
- Spearheaded infrastructure for **background thread process scheduling** of expensive metadata generation for the SQLite database, such as indexing and segmentation for large notes with up to **10K handwritten strokes**.

Software Engineer | Google Docs Web Language Intelligence | Google India, Bangalore ***Aug 2020 - Aug 2022***

- Architected multi-language web client **spellcheck for encrypted documents** via an on-device model in a parallel worker thread hosting a C++ model as a **WebAssembly binary** in a Java + JS client, triggering on **70%+ docs** with an **~80% acceptance**, and pushing for robust debugging tools from the **Emscripten** open source repository.
- Deployed intelligent **writing style suggestions for Google Docs** affecting **50%+ docs**, with an **acceptance rate of ~50%**, driving the development and hosting of the backing learning model, the client framework, and the UI.

Internships

- Teaching Assistant for *Natural Language Processing*, Indian Institute of Technology Madras ***Jan - May 2020***
- Software Engineering Intern, Google Docs Team, Google India, Bangalore ***May - July 2019***
- Research Intern (Natural Language Processing), Adobe Research, Bangalore ***May - July 2018***
- Research Intern (Natural Language Processing), Indian Institute of Science, Bangalore ***May - July 2017***

SKILLS

- Languages: Java, C++, Python, C#, JavaScript (Closure), HTML/CSS
- Systems & Web: WebAssembly (WASM), Android SDK, Client Architecture, Data Compression, Serialization
- Tools & Performance: Chrome DevTools, Performance Profiling, Memory Optimization, Source Maps

EDUCATION AND RESEARCH

Master of Technology and Bachelor of Technology (Five-Year Degree) in Computer Science ***Graduated May 2020***
Indian Institute of Technology Madras, India *CGPA: 8.78 out of 10*
Master's Thesis on Using Transformers for Paraphrasing Diversity, affiliated with *Carnegie Mellon University*